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Telematics and Informatics 35 (2018) 293–303

Contents lists available at ScienceDirect

Telematics and Informatics

journal homepage: www.elsevier.com/locate/tele

What drives live-stream usage intention? The perspectives of flow, entertainment, social interaction, and endorsement

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ARTICLE INFO

Keywords:
Live-stream
Flow
Entertainment
Social interaction
Endorsement

ABSTRACT

Live-stream is the real-time audio and video transmission of an event over the Internet. Since a live-stream can also be interactive, it has some of the characteristics of a small community. People often become addicted to television, movies, or video blogs for the sake of entertainment. They can also become blind and obsessive about worshiping popular idols. Moreover, their willingness to use media is affected by their own perception of the media or their expectation of the benefits. This study therefore proposes four key factors: flow, entertainment, social interaction, and endorsement. We use path analysis to test their relationships with attitude, perceived value and watching intention to understand the influential factors and channels that induce audiences to watch live-streaming events.

Analysis of 313 valid responses to a survey questionnaire indicate that approximately 65% of Taiwanese people watch live-streams via social network sites. In general, they watched live-streams simply because they liked to do so. The main reason behind their favorable impression of live-streaming was that live-streams made people happy and relieved stress. Secondly, people were attracted by the charm of the streamers, but the attraction differs slightly by gender. Females developed a more favorable impression of a live-stream when they admired the streamer. Males, on the other hand, developed a more favorable impression of a live-stream when they interacted with the streamer or other audience members. Young people tended to like live-streaming because it gave them a sense of flow, while older people were not affected by flow. This study contributes to the development of live-stream services and future studies on live-streaming.

1. Introduction

Live-streaming is a medium that records and broadcasts in real-time. Transmission is accomplished using one or more kinds of communication technology that allow images and sound to be immediately sent from one location to another, enabling audiences to feel like they are present at the event. Since its establishment in 2011, Twitch has become the world's largest live game platform. As of 2016, Facebook and YouTube have also successfully provided live-stream services. The applications of live-streaming have become more diverse over time. In addition to such common applications as live videogame play, live sporting events, and other live events, live-streams on hot topics, product experiences, talent shows, life sharing, and erotic arousal have been broadcast by ordinary people (i.e., the general public). There are even more unexpected applications waiting to be explored. According to InsightXplorer (2016), 93.8% of Taiwanese people were aware of live-stream services, and 41.3% of them had watched a live-stream. In 2016, the TWNIC

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https://doi.org/10.1016/j.tele.2017.12.003
Received 31 October 2017; Received in revised form 21 November 2017; Accepted 3 December 2017
Available online 05 December 2017
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(Taiwan Network Information Center) also pointed out that 21.9% of Taiwanese people who use the mobile internet also used a video live-stream platform. The Ericsson Mobility Report (Ericsson ConsumerLab, 2016) found that, in the U.S., 65% of heavy users of mobile apps also used a live-stream app; and that this same usage ratio was also very close South Korea.

The rapid development of live-stream services has changed people's viewing behaviors. Chinaidr.com (2017) pointed out that live-streaming is likely to become part of the audio and video mainstream as well as the social mainstream in the future. Most past studies on live-streaming have focused on Twitch (Sjöblom and Hamari, 2017; Vosmeer et al., 2016; Nascimento et al., 2014) and have put less emphasis on other live-streams. Over the years, developments in technology and the internet have ensured that low levels of usefulness and ease of use are no longer obstacles to users' willingness to use live-streaming; therefore, there is no need to further explore those aspects. The content of live-streaming is also no longer limited to live videogame play: a wider variety of live-streams has been developed. As new applications of live-streaming have appeared, audiences' viewing behaviors are likely to have changed. Correspondingly, the current affecting factors may be different from those that have been explored in past studies. This study addresses the following research questions:

- What factors influence audiences to watch live-streams? How do these factors correlate, if at all?
- Do audiences watch live-streams because they like the activity or because the sense of benefits derived from live-streaming triggers audiences' willingness to watch?

2. Derivation of theory and hypotheses

2.1. Flow theory

Flow is a state of mind in which people find themselves extremely devoted to their current activity. Flow theory was first proposed by Csikszentmihalyi (1975) to explain how complete engagement in an activity induces a special mental state in which the level of concentration allows the participant to filter out other unrelated perceptions. Ghani and Deshpande (1994) believed that the state of flow is characterized by concentration and enjoyment. That is, a state of flow exists when people maintain concentration and perceive psychological enjoyment while engaging in an activity. Wu (2009) studied the motivations of social platform users and noted that while platform developers should design for ease of use and usefulness, they should also enhance users' ability to reach a mental state of flow, inducing users to form a positive attitude so they will continue to use the platform.

This study defines flow as the condition in which the people watching a live-stream is so focused on the content of that live-stream that they are not easily disturbed by the outside world; they forget the time and even forget things they ought to do. Past studies on social platforms (Hung et al., 2016), instant messaging (Lu et al., 2009), and websites (Van Noort et al., 2012) have pointed out that flow affects attitude, thereby impacting the intention to use the particular system or function. Furthermore, scholars (Hung et al., 2010) have noted that flow affects the perceived value of a website, thereby affecting customer loyalty, patronage intension, and donations. Therefore, this study proposes the following two hypotheses:

H1a. "Flow" positively impacts audiences' "attitude" toward a live-stream.

H2b. "Flow" positively impacts audiences' "perceived value" of a live-stream.

2.2. Entertainment

Entertainment is a kind of activity that causes the audience to maintain concentration and interest. Its purpose is to make the audience feel happy or interested, or to relieve pressure by offering a temporary escape from reality and allowing them to put their worries behind them. Bosshart and Macconi (1998) defined the experience of entertainment as pleasure, thrill, relaxation and diversion. A live-stream is, by nature, a type of audio and video medium, and, thus, is characteristically entertaining. Related studies on usage and satisfaction theory have noted that audiences hope to use media for entertainment and to relieve stress (McQuail, 2010).

Past studies have found that entertainment positively impacts attitude, thereby impacting the willingness to recommend and the intention to use a given social platform (Curras-Perez et al., 2014). Kim et al. (2013) noted that hedonic motivation (fun, enjoyment, and pleasure) impacts perceived value, thereby impacting smartphone addiction. Studies regarding websites have found that entertainment impacts perceived value, thereby impacting loyalty to a given website (Kim and Niehm, 2009). The interestingness of the website impacts flow, which in turn impacts customer satisfaction and purchase intention (Hsu et al., 2012). Therefore, this study proposes the following three hypotheses:

H2a. "Entertainment" positively impacts audiences' "attitude" toward watching a live-stream.

H2b. "Entertainment" positively impacts audiences' "perceived value" of watching a live-stream.

H2c. "Entertainment" positively impacts audiences' "flow" when watching a live-stream.

2.3. Social interaction

Social interaction refers to a process in which people communicate and work or play together in society. Jaegher et al. (2010)

defined social interaction as a complex phenomenon involving different dimensions such as verbal and nonverbal behaviors, varying contexts, number of participants, imposition of reciprocal and joint activity, and expression of oneself with specific timing. Various types of live-stream services generally offer interactive features such as text chat rooms or donations, thus facilitating communication and interaction between live-stream participants. Therefore, this study defines social interaction as exchanges between the audience and the streamer, or exchanges between members of the audience watching the same live-stream.

Studies on social platform recommendations and usage behaviors have shown that social interaction impacts a user's attitude toward a given social platform, which in turn impacts recommendations and usage intention (Curras-Perez et al., 2014; Chiang, 2013). Furthermore, studies on smartphone addiction (Kim et al., 2013) and instant messaging (Zhang et al., 2017) have found that social interaction impacts perceived value, thereby impacting usage intention. Therefore, this study proposes the following two hypotheses:

H3a. "Social interaction" positively impacts audiences' "attitude" toward watching a live-stream.

H3b. "Social interaction" positively impacts audiences' "perceived value" of a live-stream.

2.4. Endorsement

Personal endorsements are affective rather than cognitive, and, thus, may occur in a conscious or unconscious mental state. Ohanian (1991) argued that endorsement involves three characteristics: attractiveness, trustworthiness and expertise. When a streamer reaches a certain level of popularity, s/he becomes well-known and hence can have the same kind of effect as other celebrities, the so-called celebrity endorsement effect. Such popular streamers are, in fact, called internet celebrities. In this study, endorsement refers to the condition in which the people watching the live-stream are attracted by the charm of the streamer; they agree with the streamer's behavior or attitude and, hence, desire to know the streamer's opinion or feelings.

Celebrity endorsements have been shown to increase the good impression of a product or brand in the minds of customers. Most past studies have focused on the advertising effectiveness of celebrity endorsements, and have explored the three characteristics of endorsement separately (Wang et al., 2017; Wei and Wu, 2013), or have deemed these three characteristics to be a single factor (Chan et al., 2013; Shafiq et al., 2011). Shafiq et al. (2011) and Chi et al. (2011) noted that a celebrity endorsement positively impacts perceived value, thereby enhancing the customer's purchase intention. In a study of social platforms (Kim and Song, 2016), celebrity self-disclosure was found to impact fans' perception of parasocial interaction. Parasocial interaction (PSI) is a performer's one-way influence on the audience, which allows the audience to hold the fantasy or feeling of having a face-to-face conversation with the performer via media, to see the performer as their friend, and then try to be close to the performer in reality. Therefore, this study proposes the following three hypotheses:

H4a. "Endorsement" positively impacts audiences' "attitude" toward watching a live-stream.

H4b. "Endorsement" positively impacts audiences' "perceived value" of a live-stream.

H4c. "Endorsement" positively impacts audiences' "social interaction" with a live-stream.

2.5. Attitude

Attitude refers to people's opinion or perception of something, i.e., the positive or negative emotion associated with something that is liked or disliked. In this study, attitude refers to people simply holding a favorable attitude toward a live-stream without regard to the benefits or value of the experience, i.e., an unspoken positive disposition toward the live-stream.

Studies on the impact of attitude on intention in the contexts of social platform usage (Hung et al., 2016; Huang and Shiau, 2015; Curras-Perez et al., 2014; Wu, 2009), instant messaging software usage (Lu et al., 2009), and product endorsement (Wang et al., 2017; Chiua and Lin, 2012) have empirically supported the notion that attitude is an important factor influencing behavioral intention. Therefore, this study proposes the following hypothesis:

H5. "Attitude" positively impacts the audience's "watching intention" toward a live-stream.

2.6. Perceived value

Perceived value is defined as a person's sense that certain objects or activities can bring benefits. In this study, perceived value refers to the benefits audiences believe live-stream media will bring: a joyful mood, a sense of enjoyment, a deeper understanding of a person with the same interest, or the enhancement of personal knowledge. As long as the audience believes a live-stream can bring value, though that value may not yet have been obtained, that belief is consistent with the definition of perceived value of a live-stream in this study.

Previous studies (Hsu and Lin, 2016; Cocosila and Igonor, 2015; Kim et al., 2013; Yu et al., 2013) have divided perceived value into hedonic value, utilitarian value, and social value, and have explored related behaviors on social websites or smartphones. Moreover, certain studies (Chi et al., 2011; Shafiq et al., 2011) have synthesized these three perceived values as a single factor in their explorations of advertising endorsement or live videogame play. Therefore, this study proposes the following hypothesis:

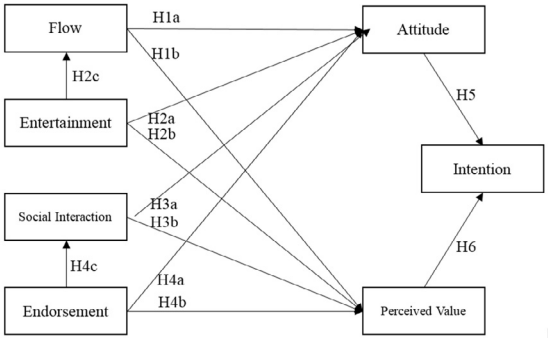


Fig. 1. Research Model.

H6. "Perceived value" positively impacts audience's "watching intention" toward a live-stream.

2.7. Research model

Based on the above theoretical exploration, we consider live-streaming to contain three factors: entertainment, social interaction, and flow. Furthermore, streamers have the charm or expertise to earn the audience's endorsement so we propose endorsement as a factor. People's own opinion of a live-stream, along with their value evaluation, may cause them to believe that the live-stream is beneficial. Therefore, this study proposes the factors of attitude and perceived value to investigate the watching intension of live-stream audiences. Our research model is shown in Fig. 1.

3. Research method

3.1. Descriptive statistics

Our questionnaire used 5-point Likert scales, and all questions were designed using topics raised by past scholars. Since the objective of this study was to explore the factors that influence audiences' watching intention, we modified and adjusted the questions to fit the scenario of watching a live-stream. The questions were reviewed and examined by experts and scholars. Past studies (Kim and Yoo, 2016; Kezer et al., 2016; Shen and Khalifa, 2010) have found gender and age differences in the use of social platforms. Hence, this study designed related items for subsequent group analysis. Data collection was accomplished via an online questionnaire. We set up a questionnaire using Google Forms and distributed it on social media platforms. The questionnaire was

Table 1
Statistics of Sample Structure.

Measure	Items	Frequency	Percent
Gender	Female	178	56.9%
	Male	135	43.1%
Age	Under 18 years old	3	0.9%
	18–25 years old	109	34.8%
	26–35 years old	90	28.8%
	36–45 years old	90	28.8%
	Over 46 years old	21	6.7%
Frequency	Rarely	132	42.2%
	Sometimes	137	43.8%
	Usually	44	14.0%
Kind	Game	78	24.9%
(Choice 1–3 items)	Sport	66	21.1%
	News Discussion	91	29.1%
	Product Experience	95	30.4%
	Talent Show	117	37.4%
	Activity Involved	57	18.2%
	About Life	123	39.3%
	Sex & Desire	10	3.2%
	Other	17	5.4%
Platform	Typical Live Platform	48	15.3%
	SNS	204	65.2%
	Vlog	61	19.5%
	Other	0	0.0%

Table 2
Results of Reliability and Validity Analysis.

Constructs & Item		Cronbach's α	Factor Loading	CR	AVE
Flow	FLO1	0.835	0.851	0.8697	0.6268
(FLO)	FLO2		0.855		
	FLO3		0.706		
	FLO4		0.744		
Entertainment	ENT1	0.831	0.762	0.8341	0.6269
(ENT)	ENT2		0.850		
	ENT3		0.760		
Social Interaction	SI1	0.873	0.759	0.8795	0.6463
(SI)	SI2		0.817		
	SI3		0.811		
	SI4		0.827		
Endorsement	EDS1	0.870	0.834	0.8697	0.6270
(EDS)	EDS2		0.852		
	EDS3		0.791		
	EDS4		0.679		
Attitude	ATT1	0.832	0.736	0.7529	0.5070
(ATT)	ATT2		0.787		
	ATT3		0.600		
Perceived Value	PV1	0.775	0.751	0.7175	0.5595
(PV)	PV2		0.745		
Intention	INT1	0.836	0.814	0.7494	0.5999
(INT)	INT2		0.733		

distributed from May 5th to May 24th, 2017. A total of 376 copies were distributed and collected. After initial filtering, there remained a total of 313 valid questionnaires (rate of return: 83.2%). The sample structure is shown in Table 1. This sample was comprised of more females (56.9%) than males (43.1%), and most respondents were between the ages of 18 and 45 (92.4%). Most respondents (86%) rarely or occasionally watched live-streams, and about 65.2% of the respondents who watched live-streams did so via social media websites.

3.2. Reliability and validity

This study adopted Nunnally's (1978) recommendation that the value of Cronbach's α should be greater than 0.7 to meet the standard of reliability. Table 2 shows that the reliability values of this study exceeded the threshold, indicating good internal consistency. Validity analysis for this study revealed a KMO value of 0.889, exceeding Kaiser's (1974) recommended value of at least 0.8. Hence, this study is suitable for factor analysis. According to Fornell and Larcker (1981), the factor loading of each dimension should be greater than the threshold value of 0.5, the construct reliability (CR) of each dimension should be greater than the threshold value of 0.7, and the average variance extracted (AVE) should be greater than the threshold value of 0.6. All data for this study met or exceeded these standards, ensuring construct validity.

Discriminant validity was measured using the Pearson Correlation Coefficient. The matrix of correlation coefficients in Table 3 shows that the square root of the AVE for each dimension is greater than the correlation coefficients between the dimensions. Therefore, each dimension of this study is sufficiently discriminant from the others, indicating good discriminant validity.

3.3. Model goodness of fit

This study used AMOS as the tool for structural equation model analysis to evaluate the model's goodness of fit indicators. The absolute fit measure index conformed to the four indexes (i.e., x^2/df, GFI, AGFI and RMSEA) proposed by Hair et al. (2010) and Bagozzi and Yi (1988). Moreover, the incremental fit index conformed to the three indexes: IFI, NNFI, and CFI (Hair et al., 2010). In addition, PNFI, PGFI, and PCFI all conformed to the recommended values (Bagozzi and Yi, 1988; Mulaik et al., 1989; Hair et al.,

Table 3
Analysis of Discriminant Validity.

	FLO	ENT	SI	EDS	ATT	PV	INT
FLO	**0.792**	–	–	–	–	–	–
ENT	0.365	**0.792**	–	–	–	–	–
SI	0.213	0.331	**0.804**	–	–	–	–
EDS	0.272	0.280	0.513	**0.792**	–	–	–
ATT	0.290	0.555	0.384	0.411	**0.712**	–	–
PV	0.342	0.471	0.354	0.366	0.655	**0.748**	–
INT	0.348	0.501	0.357	0.361	0.661	0.556	**0.775**

Table 4
Test of Model Fit.

Indexes		Suggested	Actual	Result	References
Absolute Fit Measures	x^2/df	< 3	2.608	Satisfied	Hair et al. (2010)
	GFI	> 0.8	0.865	Satisfied	Bagozzi and Yi (1988)
	AGFI	> 0.8	0.827	Satisfied	Bagozzi and Yi (1988)
	RMSEA	< 0.08	0.072	Satisfied	Hair et al. (2010)
Incremental Fit Measures	IFI	> 0.9	0.918	Satisfied	Bagozzi and Yi (1988), Hair et al. (2010)
	NNFI(TLI)	> 0.9	0.903	Satisfied	
	CFI	> 0.9	0.917	Satisfied	
Parsimonious Fit Measures	PNFI	> 0.5	0.745	Satisfied	Bagozzi and Yi (1988), Mulaik et al. (1989), Hair et al. (2010)
	PGFI	> 0.5	0.674	Satisfied	
	PCFI	> 0.5	0.782	Satisfied	

2010). In general, our model showed acceptable goodness of fit. Table 4 shows the suggested values, measured values and analysis results of each index.

4. Study results

4.1. Results of hypothesis testing

For hypothesis verification, we used AMOS statistical software to estimate the standardized path coefficient (β value) and level of significance of the hypotheses. The results are shown in Fig. 2. Among the 12 hypotheses of this study, only H1a was disproved, i.e., the impact of flow on attitude was found to be insignificant. The rest of the hypotheses were supported. In summary, flow positively impacted perceived value; entertainment significantly impacted attitude; perceived value and flow both reached the level of significance at $p < .001$; social interaction had a positive impact on attitude and perceived value; and endorsement positively impacted attitude, perceived value and social interaction. Regarding the factors affecting watching intention, the effect of attitude was greater than that of perceived value. Among the factors affecting attitude, entertainment was found to have the highest impact. Entertainment was also found to have the highest impact on perceived value. Therefore, it can be deduced that entertainment positively impacted audiences' attitude toward the live-stream, which, in turn, positively impacted watching intention. Thus, in this study, entertainment was found to be the main channel affecting audiences' intention to watch a live-stream.

4.2. Gender differences

Past studies have found behavioral differences in social platform usage among females and males (Kim and Yoo, 2016; Shen and Khalifa, 2010). Females have been found to use social platforms more frequently than do males, and males have been shown to

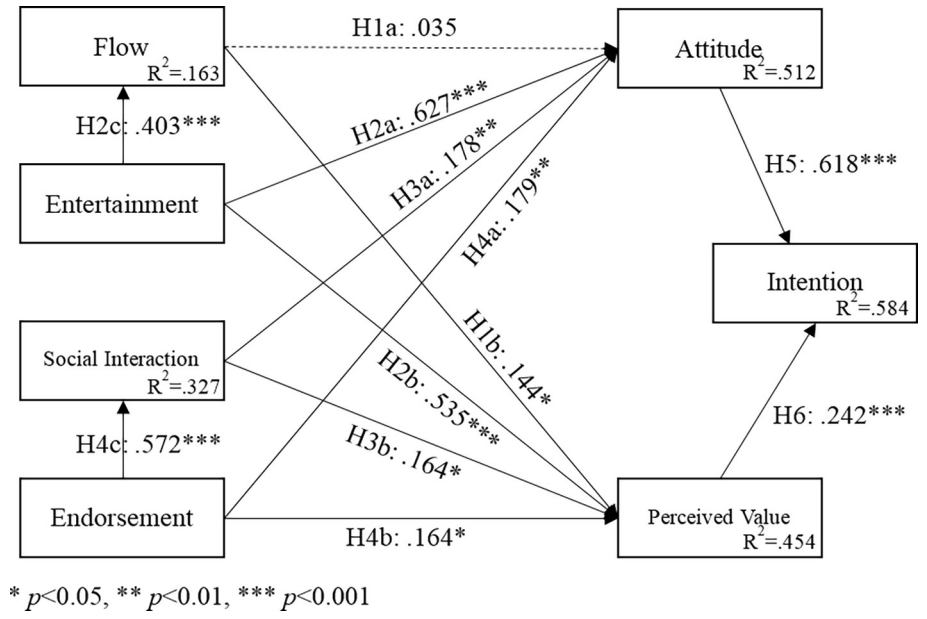


Fig.2. Result of Path Analysis of the Research Model.

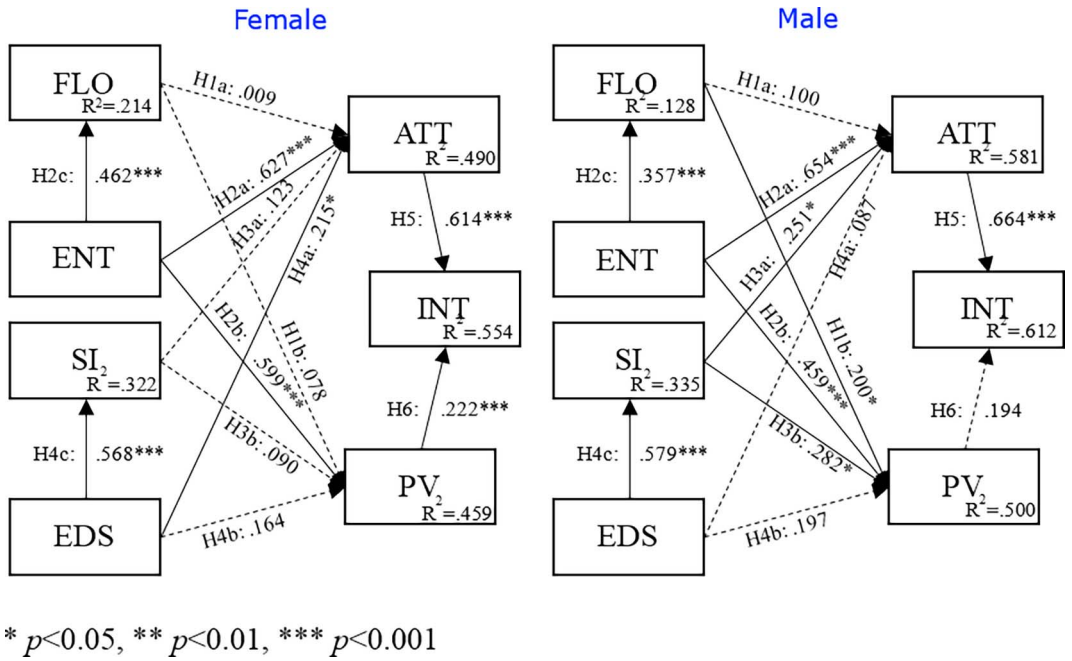


Fig.3. Result of Path Analysis of Research Model (Gender Difference). (Note: FLO, ENT Entertainment, SI: Social Interaction, EDS: Endorsement, ATT: Attitude, PV: Perceived value, INT: Intention).

consider using a social platform to be a waste of time. Therefore, this study further analyzed males and females separately.

Fig. 3 shows the results of path analysis in regards to gender differences. The data indicates that attitude had the greatest impact on watching intention when the live-stream audience was female, and attitude was most strongly influenced by entertainment (even more so than endorsement). Furthermore, for females, perceived value impacted watching intention, and entertainment had the strongest impact on perceived value. We also found that entertainment impacted flow, and endorsement impacted social interaction for females. For the male audience, live-stream watching intention was found to be impacted by attitude, and less affected by perceived value. Also for men, entertainment and social interaction were shown to positively affect attitude, while perceived value is impacted by flow, entertainment, and social interaction. On the other hand, endorsement was not found to have any significant effect on the male audience's attitude and perceived value, while endorsement was found to have a positive effect on social interaction. Entertainment was also found to affect flow for male audiences.

This study found that the main factor affecting watching intention for both male and female audiences is attitude, and entertainment is the influential factor with the greatest effect on attitude. Four major gender differences were found in regards to live-streams. Firstly, while perceived value has an impact on females' watching intention, it has no impact on males' watching intention. Secondly, social interaction was found to have an impact on attitude and perceived value for males, but not for females. Endorsement had a direct impact on attitude for females, whereas social interaction had an indirect impact on attitude for males. Finally, flow impacted perceived value for males, but had no consequent impact on watching intention. In contrast, flow did not impact perceived value for females.

4.3. Age differences

According to iResearch (2016) and INSIDE (2016), 80% of the live-stream audiences from China and Taiwan—as well as those of the world's largest streaming platform for gamers, Twitch—are under the age of 36. Yahoo's Digital Marketing Column (2016) stated that young people are more interested in live-streams, and the live-stream audience participation is highest among young people under the age of 24 (32%), followed by the age group of 34 to 35-year-olds (31%). In a survey regarding audio- and video-related websites, InsightXplorer (2016) found that the time spent and number of visits of the 15 to 24-year-old group account for approximately 32% of the total number of people in Taiwan. Since 25 is generally the age at which Taiwanese youth enter the workplace, this study divided the sample into two age groups for analysis purposes: those age 25 years old and younger, and those age 26 years old and over.

In regards to those age 25 and under, Fig. 4 shows that watching intention is impacted by attitude and perceived value. Flow and entertainment both positively impact attitude and perceived value, while social interaction impacts neither. Endorsement was found to impact attitude but not perceived value. Our proposals that entertainment impacts flow and endorsement impacts social interaction are supported by the data.

Among respondents 26 years old and over, watching intention is impacted by attitude and perceived value. Flow is shown to have no significant impact on either attitude or perceived value, whereas entertainment impacts both significantly. Social interaction

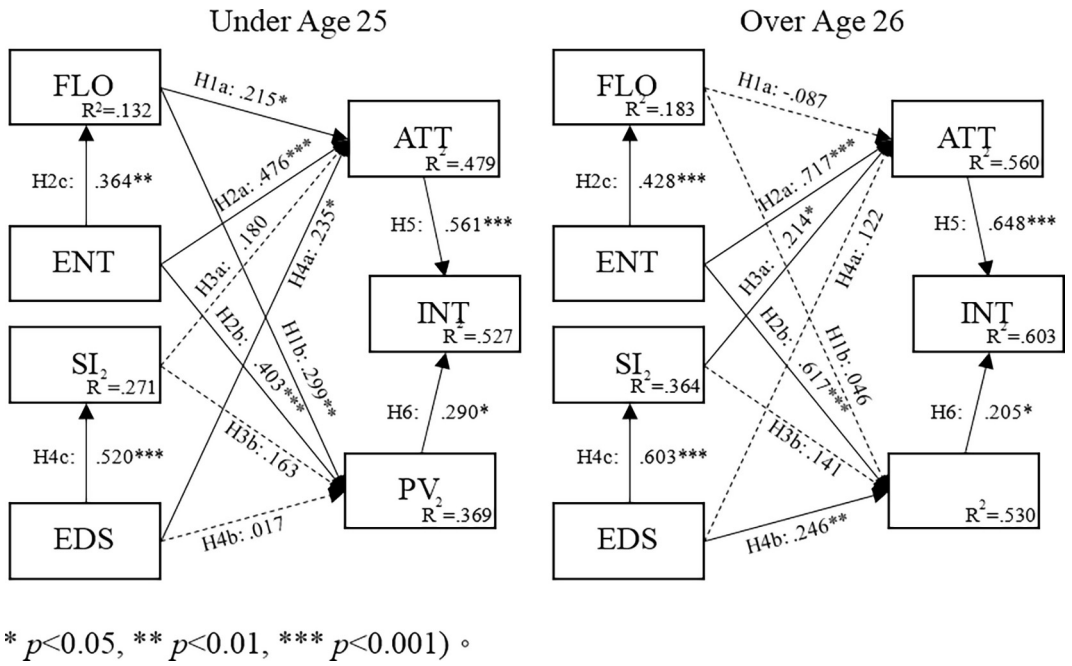


Fig. 4. Result of Path Analysis of Research Model (Age Difference). (Note: FLO, ENT Entertainment, SI: Social Interaction, EDS: Endorsement, ATT: Attitude, PV: Perceived value, INT: Intention).

impacts attitude but not perceived value. In contrast, endorsement impacts perceived value but not attitude.

For both age groups (25 and under, and 26 and over), the main factor affecting viewing intention is attitude; and attitude is most affected by entertainment. The major difference between the two age groups is the effect of endorsement on attitude and perceived value. For the younger group, endorsement impacts attitude but not perceived value. In contrast, for the older group, endorsement affects perceived value but not attitude. Secondly, for the young group, flow impacts perceived value, whereas perceived value is not significantly impacted by flow for the older group. Finally, social interaction affects the older group's attitude toward a live-stream, but the effect of social interaction on attitude was not significant among the younger group.

5. Conclusion

5.1. Research conclusion

The results of this study show that, regardless of whether the sample is taken as a whole or divided by gender or age, the major channel affecting an audience's watching intention is via entertainment's effect on attitude, which consequently affects watching intention. Next, attitude toward a live-stream is directly affected by endorsement and indirectly affected by social interaction, thereby affecting watching intention. Flow affects watching intention via perceived value. However attitude is more influential than perceived value, and male audiences' watching intention is unaffected by perceived value. Thus, flow has little effect on watching intention, making flow the least influential factor.

Flow had a significant impact on attitude for only one group: those age 25 and under. It did not significantly impact attitude for males in general, females in general, those 26 years old and over, or the sample as a whole. This finding was inconsistent with those of previous studies on social platforms (Hung et al., 2016; Huang and Shiau, 2015). One possible reason for this is that the richness of live-streaming is low compared to concerts or fan meetings which are often attended by fans. If flow was the reason why audiences watched media, those audiences had other options (e.g., TV series, movies, Vlogs and more). Moreover, audiences usually multitask when watching a live-stream (InsightXplorer, 2017). Statistics show that 63.5% of audiences perform other tasks while watching a live-streaming event, which might explain why live-stream audiences do not find that watching a live-stream provides any sense of flow. The reason why flow did not significantly impact perceived value for females might be related to the type of live-streams being watched. According to this study's sample, the most common type of live-stream watched by females is life sharing (42.1%) while males most often watch live videogame play (41.5%). Watching life sharing live-streams does not require the same kind of concentration required for live games, for which the audience must focus on the content in order to learn game skills. Those 25 and under might be more dedicated to live-streaming events, in general. People who are over the age of 25 may already have entered the workplace. Even during their spare time, they are often distracted by housework and work tasks. Achieving a state of flow becomes difficult, reducing the impact of flow to insignificance. Live-streams on social websites are temporary interstitial events that allow users to temporarily put aside their tasks; thus flow does not have a significant effect on perceived value.

Entertainment has a positive and significant impact on attitude, perceived value, and flow, a finding which is consistent with

those of past studies on social websites (Curras-Perez et al., 2014), general websites (Hsu et al., 2012; Kim and Niehm, 2009), and addiction to smartphones (Kim et al., 2013). Compared to flow, social interaction and endorsement, entertainment had a higher degree of impact on attitude and perceived value. Though its impact was not as high as that of entertainment, social interaction also had a positive and significant impact on attitude and perceived value. That is, an increase in the degree of interaction between the audience and streamer or between audience members also increased the audiences' positive attitude toward the live-stream and enhanced the perceived value. This was consistent with the findings of past studies on social platforms (Curras-Perez et al., 2014; Chiang, 2013). The greatest gender difference regarding the impact of social interaction on attitude was found in the male group. This finding is inconsistent with past findings that females and young people more often use social platforms to communicate and interact with others (Kim and Yoo, 2016; Shen and Khalifa, 2010). This might be related to the fact that male streamers are completely outnumbered by female streamers, who are more likely to attract interactive attention from male audiences. Furthermore, young people might be dedicated to live-streaming and ignore the existence of text-only chat rooms, resulting in the insignificant impact of social interaction on attitude and perceived value.

In this study, endorsement was found to have a positive and significant impact on attitude and perceived value. That is, the higher the degree to which the audience recognizes the streamer, the more positive the attitude of the audience toward the streamer. This increases the audience's sense of the worthiness of the streamer, encouraging the audience to interact during the live-stream. This is consistent with past findings on product or service endorsements (Wang et al., 2017; Shafiq et al., 2011; Chi et al., 2011). When the audience's level of endorsement was high, the audience was more likely to interact with the streamer during the live-stream. It is possible that males are more rational when dealing with idols, so the direct impact of endorsement on attitude, like perceived value, was insignificant. Endorsement impacted attitude only through interaction with the streamer. Those age 25 and under are generally less experienced in social circles and might be more likely to appreciate the surrounding context because they admire the streamer, so endorsement affected attitude for this group. People over age 25 might subconsciously calculate gains and losses, thus being more impacted by the perceived value of the live-stream.

The impact of attitude on watching intention was found to be significant, consistent with previous empirical results regarding similar media such as social platforms (Hung et al., 2016; Huang and Shiau, 2015; Curras-Perez et al., 2014; Wu, 2009) and instant messaging software (Lu et al., 2009). The more benefits people believe they can gain from a live-stream, the stronger the intention to continue to watch the live-stream. This finding is consistent with past studies on smartphones (Kim et al., 2013), websites (Hung et al., 2010; Soltani and Gharbi, 2008), and advertising endorsements (Chi et al., 2011). Group analysis revealed one notable exception: for males, watching intention is unaffected by perceived value. For all other groups (i.e., females, younger respondents and older respondents), the effects of influential factors on watching intention are quite consistent. Shen and Khalifa (2010) found that, compared to females, males tend to consider using a social platform to be a waste of time. This might provide some explanation as to why perceived value had an insignificant impact on males' watching intention. In other words, male audiences might consider watching a live-stream to be a waste of time, thus diminishing the perceived value, resulting in an insignificant impact on watching intention.

5.2. Implications to Management and practice

Past academic research has often focused on live videogame play. This narrow context is inadequate for today's diverse applications of live-streaming. Moreover, in our exploration of how attitude or perceived value affects an audience's intention to watch a live-stream, we used factors that apply to psychological needs or substance usage, i.e., flow, entertainment, social interaction, and endorsement. We found that entertainment affects attitude, which, in turn, affects watching intention. Thus, entertainment is the main factor affecting an audience's intention to watch a live-stream.

Based on the findings of this study, we recommend that live streaming platforms establish a small-scale professional team, and broadcast well-designed programs as live-streams in order to increase the entertainment value. Alternatively, we recommend that live streaming platforms nurture ordinary people who have the talent to be streamers and become internet celebrities. Giving streamers proper guidance as to how to properly interact with the audience will help to enhance audiences' positive attitude toward the live-stream and increase their intention to watch the live-stream. To make social interaction more frequent and more effective, the platform itself should focus on developing the interactive function so that streamers and audiences have good interaction tools. This should help increase the activity of live-stream audiences. If most of the platform's users are female, we suggest giving priority to employing attractive, professional, and trustworthy streamers to boost females' positive attitude toward the live-stream. If the platform's user base is male-dominated, we suggest increasing the number of female streamers and diversifying the live-streams. We also recommend that streamers maintain their enthusiasm and interact frequently with the audience during the live-stream, since a favorable feeling toward the live-stream is induced indirectly via interaction.

5.3. Research limitations and future research

This study employed a questionnaire which was distributed online via a social platform, so that subjects who watched live-streams on social websites had a comparative advantage. Although the results of this study are consistent with the findings of past surveys, we recommend that future studies increase the variety of sample sources in order to improve the diversity of the sample.

This study treated all types of live-streams equally. However, audiences watching different type of live-streams might behave differently. Most past studies have focused on live-gaming. Future studies should focus on the types of live-streams that were watched most frequently by this research sample, such as life-sharing, talent shows, or commodity auctions, which have recently gained

popularity. With the rapid development of live-streams, more types of content will be live-streamed, and the audience for each type will grow, providing a suitable environment for quantitative analysis for live-streams that are not live-games.

Acknowledgements

The author would like to thank the Ministry of Science and Technology of the Republic of China for supporting this study financially (Contract No. 105-2511-S-005-001-MY3).

Appendix A

Questionnaire Items

- Flow

FLO1: When watching a live-stream, I do not realize how time passes.
FLO2: When watching a live-stream, I often forget the work I must do.
FLO3: Watching a live-stream gives me a temporary escape from the real world.
FLO4: When watching live-stream, I am not distracted easily by other things.

- Entertainment

ENT1: Watching a live-stream is entertaining.
ENT2: Watching a live-stream relaxes me.
ENT3: Watching a live-stream gives me pleasure.

- Social Interaction

SI1: When watching a live-stream, I can exchange and share opinions with the streamer or other audiences easily.
SI2: When I am watching a live-stream, the streamer knows I'm concerned about him or her.
SI3: When watching a live-stream, I feel closer to the streamer.
SI4: When I am watching a live-stream, the streamer provides sufficient opportunities to respond and ask a question.

- Endorsement

EDS1: I simply appreciate the streamer.
EDS2: The streamer is attractive or charming.
EDS3: The streamer is trustworthy.
EDS4: I can know the streamer’s feelings or views.

- Attitude

ATT1: I think watching a live-stream is a good idea.
ATT2: My attitude toward watching live-streams is positive.
ATT3: I like watching live-streams.

- Perceived Value

PV1: Watching live-streams provides me with a lot of enjoyment, or gives me happiness.
PV2: Given the time I need to spend doing it, watching a live-stream is worthwhile to me.

- Intention

INT1: I will watch a live-stream if time allows.
INT2: I will continue watching live-streams in the future.

Appendix B. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.tele.2017.12.003.

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Update

Telematics and Informatics

Volume 35, Issue 6, September 2018, Page 1794

DOI: https://doi.org/10.1016/j.tele.2018.01.002

Telematics and Informatics 35 (2018) 1794

Contents lists available at ScienceDirect

Telematics and Informatics

journal homepage: www.elsevier.com/locate/tele

Corrigendum

Corrigendum to "What drives live-stream usage intention? The perspectives of flow, entertainment, social interaction, and endorsement" [Telematics Inf. 35(1) (2018) 293–303]

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The authors regret there is one construct name missing in Fig. 4, and it should be named "PV".
The authors would like to apologise for any inconvenience caused.

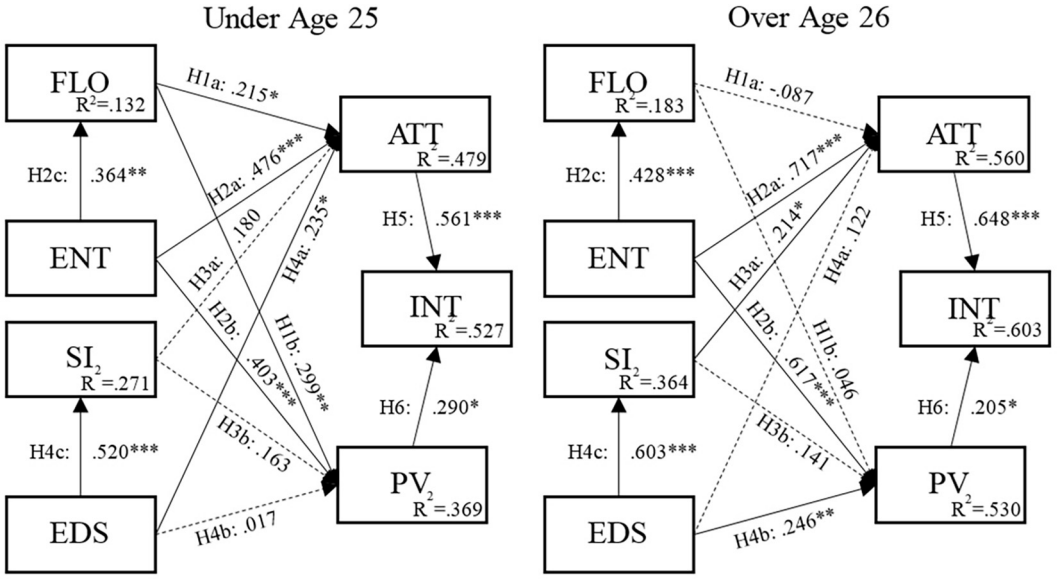


Fig. 4.

DOI of original article: http://dx.doi.org/10.1016/j.tele.2017.12.003
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https://doi.org/10.1016/j.tele.2018.01.002

Available online 22 February 2018
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